

Regression Line and Regression Equation

Regression line: The line that best fits a set of data points according to the least-squares criterion.

Regression equation: The equation of the regression line.

$$y = b_0 + b_1x$$

Regression Equation

The regression equation for a set of n data points is $\hat{y} = b_0 + b_1x$, where

$$b_1 = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad b_0 = \frac{1}{n}(\Sigma y_i - b_1 \Sigma x_i) = \bar{y} - b_1 \bar{x}.$$

Age and Price of Orions In the first two columns of Table 14.5, we repeat our data on age and price for a sample of 11 Orions.

- Determine the regression equation for the data.
- Graph the regression equation and the data points.
- Describe the apparent relationship between age and price of Orions.
- Interpret the slope of the regression line in terms of prices for Orions.
- Use the regression equation to predict the price of a 3-year-old Orion and a 4-year-old Orion.

Age (yr) x	Price (\$100) y
5	85
4	103
6	70
5	82
5	89
5	98
6	66
6	95
2	169
7	70
7	48

Age (yr) x	Price (\$100) y	xy	x^2
5	85	425	25
4	103	412	16
6	70	420	36
5	82	410	25
5	89	445	25
5	98	490	25
6	66	396	36
6	95	570	36
2	169	338	4
7	70	490	49
7	48	336	49
58	975	4732	326

Solution

- a. We first need to compute b_1 and b_0 by using Formula 14.1. We did so by constructing a table of values for x (age), y (price), xy , x^2 , and their sums in Table 14.5.

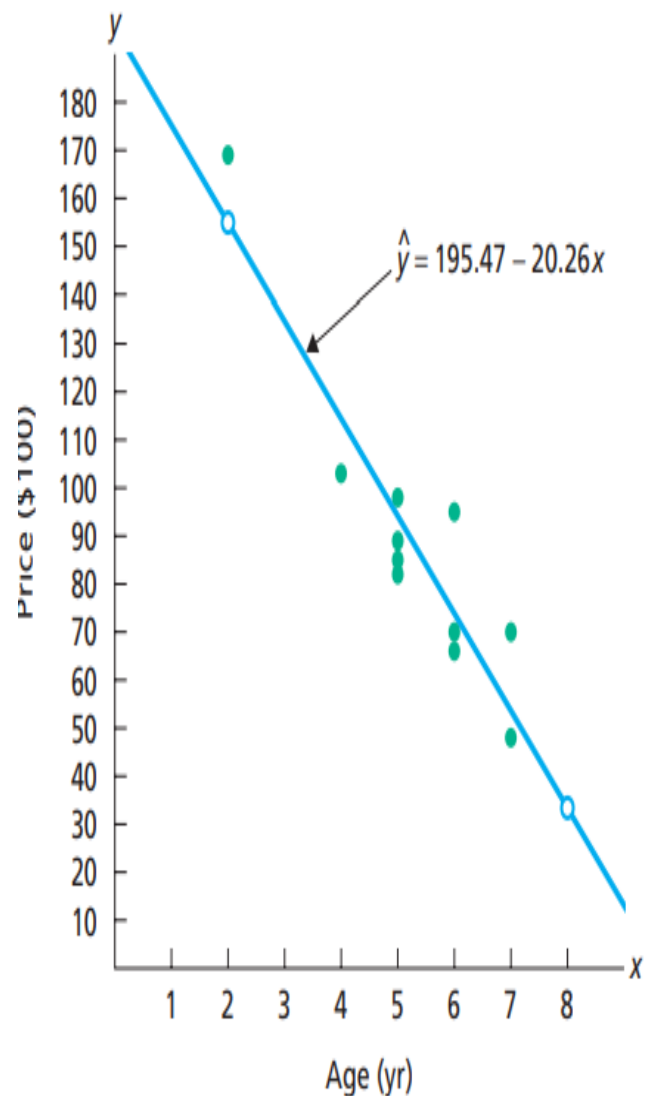
The slope of the regression line therefore is

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum x_i y_i - (\sum x_i)(\sum y_i)/n}{\sum x_i^2 - (\sum x_i)^2/n} = \frac{4732 - (58)(975)/11}{326 - (58)^2/11} = -20.26.$$

$$b_0 = \frac{1}{n}(\sum y_i - b_1 \sum x_i) = \frac{1}{11}[975 - (-20.26) \cdot 58] = 195.47.$$

- b. To graph the regression equation, we need to substitute two different x -values in the regression equation to obtain two distinct points. Let's use the x -values 2 and 8. The corresponding y -values are

$$\hat{y} = 195.47 - 20.26 \cdot 2 = 154.95 \quad \text{and} \quad \hat{y} = 195.47 - 20.26 \cdot 8 = 33.39.$$



- c. Because the slope of the regression line is negative, price tends to decrease as age increases, which is no particular surprise.
- d. Because x represents age in years and y represents price in hundreds of dollars, the slope of -20.26 indicates that Orions depreciate an estimated \$2026 per year, at least in the 2- to 7-year-old range.
- e. For a 3-year-old Orion, $x = 3$, and the regression equation yields the predicted price of

$$\hat{y} = 195.47 - 20.26 \cdot 3 = 134.69.$$

Similarly, the predicted price for a 4-year-old Orion is

$$\hat{y} = 195.47 - 20.26 \cdot 4 = 114.43.$$

```
variable,  
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1) # Reshape to 2D array  
y = np.array([2, 4, 5, 4, 5])  
  
# Create a Linear Regression model  
model = LinearRegression()  
  
# Fit the model  
model.fit(X, y)  
  
# Regression coefficients  
slope = model.coef_[0]  
intercept = model.intercept_  
  
# Display the regression equation  
print(f"Regression equation:  $y = \text{{slope:.2f}}x + \text{{intercept:.2f}}$ ")  
  
# Make predictions  
y_pred = model.predict(X)  
  
# Calculate Mean Squared Error  
mse = mean_squared_error(y, y_pred)  
print(f"Mean Squared Error: {mse:.2f}")  
  
# Plot the results  
plt.scatter(X, y, color='blue', label='Actual data')
```

```
from sklearn.linear_model import LinearRegression
```

One-liner for regression

```
model = LinearRegression().fit(X, y) # Fit model
```

14.51 Corvette Prices. The *Kelley Blue Book* provides information on wholesale and retail prices of cars. Following are age and price data for 10 randomly selected Corvettes between 1 and 6 years old. Here, x denotes age, in years, and y denotes price, in hundreds of dollars. For part (g), predict the prices of a 2-year-old Corvette and a 3-year-old Corvette.

x	6	6	6	2	2	5	4	5	1	4
y	290	280	295	425	384	315	355	328	425	325

14.52 Custom Homes. *Hanna Properties* specializes in custom-home resales in the Equestrian Estates, an exclusive subdivision in Phoenix, Arizona. A random sample of nine custom homes currently listed for sale provided the following information on size and price. Here, x denotes size, in hundreds of square feet, rounded to the nearest hundred, and y denotes price, in thousands of dollars, rounded to the nearest thousand. For part (g), predict the price of a 2600-sq. ft. home in the Equestrian Estates.

x	26	27	33	29	29	34	30	40	22
y	540	555	575	577	606	661	738	804	496

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Sums of Squares in Regression

Total sum of squares, SST : The total variation in the observed values of the response variable: $SST = \sum (y_i - \bar{y})^2$.

Regression sum of squares, SSR : The variation in the observed values of the response variable explained by the regression: $SSR = \sum (\hat{y}_i - \bar{y})^2$.

Error sum of squares, SSE : The variation in the observed values of the response variable not explained by the regression: $SSE = \sum (y_i - \hat{y}_i)^2$.

Coefficient of Determination

The **coefficient of determination**, r^2 , is the proportion of variation in the observed values of the response variable explained by the regression. Thus,

$$r^2 = \frac{SSR}{SST}.$$

Note: The coefficient of determination, r^2 , always lies between 0 and 1. A value of r^2 near 0 suggests that the regression equation is not very useful for making predictions,

Age and Price of Orions The age and price data for a sample of 11 Orions are repeated in the first two columns of Table 14.9. Use the computing formulas in Formula 14.2 to determine the three sums of squares.

Age (yr) x	Price (\$100) y	xy	x^2	y^2
5	85	425	25	7,225
4	103	412	16	10,609
6	70	420	36	4,900
5	82	410	25	6,724
5	89	445	25	7,921
5	98	490	25	9,604
6	66	396	36	4,356
6	95	570	36	9,025
2	169	338	4	28,561
7	70	490	49	4,900
7	48	336	49	2,304
58	975	4732	326	96,129

Using the last row of Table 14.9 and Formula 14.2, we can now find the three sums of squares for the Orion data. The total sum of squares is

$$SST = \sum y_i^2 - (\sum y_i)^2/n = 96,129 - (975)^2/11 = 9708.5;$$

$$SST = \sum y_i^2 - (\sum y_i)^2/n = 96,129 - (975)^2/11 = 9708.5;$$

the regression sum of squares is

$$SSR = \frac{[\sum x_i y_i - (\sum x_i)(\sum y_i)/n]^2}{\sum x_i^2 - (\sum x_i)^2/n} = \frac{[4732 - (58)(975)/11]^2}{326 - (58)^2/11} = 8285.0;$$

and, from the two preceding results, the error sum of squares is

$$SSE = SST - SSR = 9708.5 - 8285.0 = 1423.5.$$

$$SSR = \Sigma(\hat{y}_i - \bar{y})^2 = 8285.0,$$

which is the variation in the observed prices explained by the regression.

From SST and SSR , we compute the coefficient of determination, the percentage of variation in the observed prices explained by the regression (i.e., by the linear relationship between age and price for the sampled Orions):

$$r^2 = \frac{SSR}{SST} = \frac{8285.0}{9708.5} = 0.853 \quad (85.3\%).$$

Interpretation Evidently, age is quite useful for predicting price because 85.3% of the variation in the observed prices is explained by the regression of price on age.

```
from sklearn.linear_model import LinearRegression
import numpy as np

# Example data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1) # Reshape to 2D array
y = np.array([2, 4, 5, 4, 5])

# Fit the regression model
model = LinearRegression().fit(X, y)

# Get the R^2 value
r_squared = model.score(X, y)
print(f"Coefficient of Determination (R^2): {r_squared:.2f}")
```

calculate the coefficient of determination (R^2) in a single line of Python code:

```
r_squared = LinearRegression().fit(X, y).score(X, y)
```

